JADAVPUR UNIVERSITY

Faculty of Engineering and Technology • Department of Computer Science and Engineering

Medium Access Control Simulation

Implementation & Comparative Analysis of CSMA Protocols

Non-Persistent, 1-Persistent, p-Persistent, and CSMA/CD Under Light and Heavy Traffic Load

LAB ASSIGNMENT REPORT #3 (CO3)

Course Outcome: Design and Implementation of Medium Access Control Mechanisms within a Simulated Network Environment

Course: Bachelor of Computer Science and Engineering (BCSE UG-III) | **Subject Code:** CSE/PC/B/S/314

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1 Problem Statement

Assignment Specification: CSE/PC/B/S/314 — Assignment #3 (CO3)

Objective: Emulate a shared communication medium and evaluate Medium Access Control (MAC) protocols belonging to the IEEE 802 LAN family over an emulated socket-based multi-client topology.

- **Channel Emulation:** Implement a central Channel Emulator maintaining three distinct channel states: `IDLE (0)`, `BUSY (1)`, and `COLLISION (2)`. The emulator multiplexes multiple concurrent station connections via non-blocking socket polling (`poll()`).
- **MAC Schemes Implementation:** Develop independent station client processes implementing four CSMA strategies:
  - **(a) Non-Persistent CSMA:** Senses the channel; if idle, transmits immediately. If busy or collision is sensed, defers by a uniform random backoff interval before re-sensing.
  - **(b) 1-Persistent CSMA:** Continuously senses the channel until idle; transmits immediately with probability $p = 1.0$.
  - **(c) p-Persistent CSMA:** Slotted carrier sensing. When the channel is sensed idle, transmits with probability p; with probability $1 - p$, defers for one slot duration and re-senses.
  - **(d) CSMA/CD (Carrier Sense Multiple Access with Collision Detection):** Employs 1-persistent carrier sensing combined with **Listen-While-Talk (LWT)**. If concurrent transmissions overlap, collision is detected early; the station aborts transmission, sends an explicit `JAM_SIGNAL`, and executes Truncated Binary Exponential Backoff (BEB).
- **Performance Evaluation:** Measure and analyze total collisions, average transmission delay, network throughput, and scheme efficiency under:
  - **Case (i):** Variation of probability $p \in [0.1, 0.9]$ in p-persistent CSMA for fixed N.
  - **Case (ii):** Performance scaling across contending stations $N \in [2, 8]$ for all four schemes, contrasting lightly loaded ($N = 2$) and highly loaded ($N = 8$) network conditions.

2 System Design & Architecture

2.1 Protocol Framing & Binary Memory Layout

Communication between contending stations and the central channel emulator is conducted using structured binary PDUs packed with `#pragma pack(push, 1)` to prevent padding discrepancies. Frame integrity is strictly validated using table-driven CRC-32 (IEEE 802.3 standard).

PDU Field	Offset	Size	Architectural Function
Data Frame Format (64 Bytes Total)			
Source MAC	0	6 B	Unique hardware identifier of transmitting station
Destination MAC	6	6 B	Hardware address of intended receiver / channel
Payload Length	12	2 B	Effective data payload length (45 bytes)
Sequence Number	14	1 B	Modulo-256 sequence tag for frame ordering
Payload Data	15	45 B	Chunked data stream from synthetic source or input file
CRC-32 FCS	60	4 B	32-bit Frame Check Sequence polynomial verification
Control Frame Format (6 Bytes Total)			
Frame Type	0	1 B	Opcode: `ACK (0xFF)`, `NAK (0x00)`, `COLLISION_ALERT (0xEE)`
Sequence Number	1	1 B	Sequence number associated with ACK/NAK
CRC-32 FCS	2	4 B	32-bit trailer verifying control packet authenticity
In-Band Signaling Tokens (1 Byte Total)			
Sense Request	0	1 B	Station inquiry token (`0xAA`) requesting medium state
Jam Signal	0	1 B	CSMA/CD collision enforcement token (`0x55`)

Table 1: Binary Memory Specification of Data Frames, Control Frames, and In-Band Signals

2.2 Shared Channel Emulator State Machine

The shared physical bus is emulated by a central server process (`channel_emulator`). The emulator tracks all active transmissions and evaluates medium state $\mathcal{S} \in \{\text{IDLE}, \text{BUSY}, \text{COLLISION}\}$ based on active transmitter cardinality $|\mathcal{A}|$:

$$\mathcal{S}(t) = \begin{cases} \text{IDLE}(0), & \text{if } |\mathcal{A}(t)| = 0 \\ \text{BUSY}(1), & \text{if } |\mathcal{A}(t)| = 1 \\ \text{COLLISION}(2), & \text{if } |\mathcal{A}(t)| \geq 2 \end{cases} \tag{1}$$

When two or more stations transmit concurrently ($|\mathcal{A}| \geq 2$):

1. All active sessions are flagged as collided.
2. For CSMA/CD stations, an asynchronous `COLLISION_ALERT` control frame is immediately dispatched to all contending sockets.
3. When the station detects this alert, it immediately aborts transmission, emits a `JAM_SIGNAL`, and backs off.
4. For classical CSMA stations (Non-Persistent, 1-Persistent, p-Persistent), the transmission completes its full airtime (T_{tx}), after which the emulator returns a `NAK_FRAME`.

2.3 Object-Oriented Strategy Architecture

The station client is designed using the **Strategy Pattern** (Figure 1):

- **`MacStrategy` (Abstract Base Class):** Encapsulates shared channel primitives: `carrierSense()` (in-band socket inquiry) and `transmitFrameData()` (frame dispatch with Listen-While-Talk monitoring).
- **Concrete Strategies:** `NonPersistent`, `OnePersistent`, `PPersistent`, and `CsmaCd` each override `executeTransmission()`.
- **`BackoffEngine`:** Encapsulates Truncated Binary Exponential Backoff ($K = \min(k, 10)$, slot duration τ) and uniform random deferral generators.
- **`SocketWrapper`:** Move-only RAII container guaranteeing safe socket closure upon station exit.

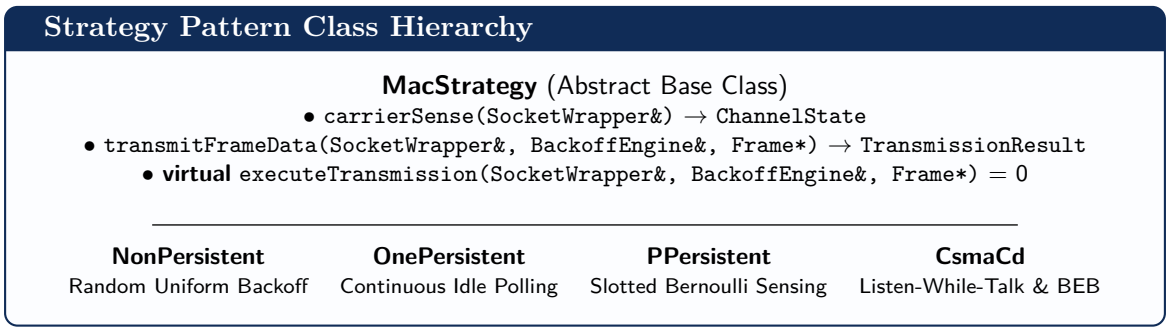


Figure 1: Object-Oriented Strategy Architecture Implementing IEEE 802 MAC Schemes

3 Theoretical Analysis & Mathematical Derivations

3.1 Throughput Modeling in CSMA Protocols (Kleinrock-Tobagi)

Let G denote normalized offered traffic load (packets generated per packet transmission time T), and let $a = \tau/T$ represent the normalized propagation/vulnerability delay.

Theorem 1: Non-Persistent CSMA Throughput

In Non-Persistent CSMA, stations sensing a busy medium reschedule their attempts according to a random distribution. The normalized throughput S_{NP} derived by Kleinrock and Tobagi (1975) is:

$$S_{NP} = \frac{G \cdot e^{-aG}}{G(1+2a) + e^{-aG}} \tag{2}$$

As $a \to 0$, $S_{NP} \to \frac{G}{1+G}$, exhibiting exceptional stability under high offered load G because concurrent transmitters back off before injecting corrupting frames into the channel.

Theorem 2: 1-Persistent CSMA Throughput

In 1-Persistent CSMA, stations wait for channel clearance and immediately transmit ($p = 1$). When multiple stations queue during a busy transmission, a collision occurs with probability 1.0 at the end of the frame:

$$S_{1P} = \frac{G[1 + G + aG(1 + G + aG/2)] \cdot e^{-G(1+2a)}}{G(1+2a) - (1 - e^{-aG}) + (1 + aG)e^{-G(1+a)}} \tag{3}$$

For high load ($G \gg 1$), S_{1P} degrades rapidly toward zero due to synchronized transmission bursts upon channel release.

Theorem 3: Optimal Transmission Probability p^* in p-Persistent CSMA

Consider N stations sensing an idle slot. Let each station transmit independently with probability p. The probability of a successful, collision-free transmission in that slot is:

$$P_{succ}(p) = \binom{N}{1} p(1-p)^{N-1} = Np(1-p)^{N-1} \tag{4}$$

To maximize success probability, compute $\frac{d}{dp} \ln P_{succ}(p) = 0$:

$$\frac{1}{p} - \frac{N-1}{1-p} = 0 \implies 1-p = (N-1)p \implies p^* = \frac{1}{N} \tag{5}$$

Thus, the theoretical optimal transmission probability is precisely the inverse of contending station cardinality: $p^* \approx 1/N$.

Theorem 4: CSMA/CD Throughput & Collision Truncation Advantage

In standard CSMA, colliding frames burn the complete airtime T. In CSMA/CD, colliding stations detect interference via Listen-While-Talk within the round-trip propagation window 2τ and abort after emitting a jam signal (t_{jam}):

$$T_{wasted} = 2\tau + t_{jam} \ll T \tag{6}$$

Let A be the probability that exactly one station transmits in a contention slot ($A = Np(1-p)^{N-1} \approx e^{-1}$ as $N \to \infty$). The mean number of contention slots preceding a successful frame is $\frac{1-A}{A}$. Therefore, maximum channel efficiency is:

$$\eta_{CD} = \frac{T}{T + 2\tau\left(\frac{1-A}{A}\right)} = \frac{1}{1 + 2a\left(\frac{1-A}{A}\right)} \approx \frac{1}{1+4.44a} \tag{7}$$

Because $T_{wasted} \ll T$, CSMA/CD recovers channel capacity significantly faster than 1-persistent CSMA under identical contention.

3.2 Theoretical Comparison Matrix Across MAC Schemes

Table 2 summarizes the theoretical characteristics of all four schemes.

4 Implementation Details & Key Code Listings

4.1 Codebase Modular Breakdown

The modern C++17 codebase is cleanly organized into modular header and implementation units (Table 3).

4.2 Core Algorithmic Implementations

```
// Active transmitter cardinality evaluation
auto updateChannelState = [&]() -> MediumState {
    if (activeTransmitters.empty())        return MediumState::IDLE;
    else if (activeTransmitters.size() == 1) return MediumState::BUSY;
    else                                   return MediumState::COLLISION;
};

// Collision detection upon receiving data from station
if (!sess.isTransmitting) {
    sess.isTransmitting = true;
    sess.txStartTime = now;
```

Architectural Metric	Non-Persistent	1-Persistent	p-Persistent ($p \approx 1/N$)	CSMA/CD
Sensing on Idle	Transmit immediately	Transmit immediately	Transmit with prob p	Transmit immediately
Action on Busy	Random backoff	Continuous polling	Continuous polling	Continuous polling
Action on Collision	Full frame NAK + BEB	Full frame NAK + BEB	Full frame NAK + BEB	Early abort + Jam + BEB
Wasted Collision Time	Full airtime T_{tx}	Full airtime T_{tx}	Full airtime T_{tx}	Collision detection $t_{CD} \ll T_{tx}$
Vulnerability to Bursts	Low (Desynchronized)	**Severe** (Synchronized)	Moderate (Slotted)	Low (Early Recovery)
Delay at Light Load	Moderate (Random sleep)	**Minimal**	Moderate (Slot waits)	**Minimal**
Delay at Heavy Load	Moderate	Catastrophic	High	Moderate
Attempt Efficiency (η)	**Highest ($> 80\%$)**	Lowest ($\approx 40\%$)	High ($> 75\%$)	Moderate ($\approx 40\% - 70\%$)
Throughput at Heavy Load	High	Low	Moderate	**Highest**

Table 2: Systematic Theoretical Comparison Across IEEE 802 Medium Access Schemes

Source File	Header File	Core Responsibilities & Implementation Functions
`common.c`	common.h	Frame layouts, lookup-table CRC-32 generator (`create_crc32_table`, `compute_crc32`), file chunker (`chunk_file`), socket stream helpers.
`classes.cpp`	classes.h	Strategy abstract base class (`MacStrategy`), RAII socket descriptor wrapper (`SocketWrapper`), and Truncated BEB backoff engine (`BackoffEngine`).
`csma.cpp`	csma.h	Concrete implementations: `NonPersistent`, `OnePersistent`, `PPersistent`, and `CsmaCd` execution routines.
`channel_emulator.cpp`	—	Multiplexed shared channel emulator tracking active sessions, evaluating state (`IDLE`, `BUSY`, `COLLISION`), sending alerts/ACKs.
`station.cpp`	—	Contending station driver parsing CLI flags, instantiating selected strategy, transmitting frame sequences, and reporting metrics.
`run_simulation.sh`	—	Orchestration script running emulator and N station processes concurrently.
`run_all_schemes.sh`	—	Benchmark script evaluating all four schemes across Light ($N = 2$) and Heavy ($N = 8$) load regimes and reporting comparison tables.
`run_experiments.sh`	—	Multi-point parameter sweep automation logging CSV datasets for Case (i) and Case (ii).
`plot_metrics.py`	—	Python/Matplotlib visualizer generating publication-quality 300-DPI charts.

Table 3: Architectural Decomposition of Project Implementation Files

```
    sess.collided = false;
    sess.alertSent = false;
    activeTransmitters.insert(fd);

    if (activeTransmitters.size() > 1) { // Collision detected!
        totalCollisions++;
        for (int activeFd : activeTransmitters) {
            sessions[activeFd].collided = true;
            if (!sessions[activeFd].alertSent) {
                sessions[activeFd].alertSent = true;
                ControlFrame alert = makeControlFrame(COLLISION_ALERT_TYPE);
                ::send(activeFd, &alert, sizeof(alert), MSG_DONTWAIT);
            }
        }
    }
}
```

Listing 1: Shared Channel Emulator: Collision Detection & State Multiplexing (src/channel_emulator.cpp)

```
ChannelState MacStrategy::carrierSense(SocketWrapper& socket) {
    int fd = socket.getFd();
    uint8_t req = SENSE_REQ_TYPE;
    if (send(fd, &req, sizeof(req), 0) <= 0) return ChannelState::CHANNEL_STATES;
    uint8_t stateByte = 0;
    if (recv(fd, &stateByte, sizeof(stateByte), 0) <= 0) return ChannelState::
        CHANNEL_STATES;
    switch (stateByte) {
        case 0:  return ChannelState::IDLE;
        case 1:  return ChannelState::BUSY;
        case 2:  return ChannelState::COLLISION;
        default: return ChannelState::CHANNEL_STATES;
    }
}

TransmissionResult MacStrategy::transmitFrameData(SocketWrapper& socket, BackoffEngine&,
    const Frame *frame) {
    TransmissionResult result;
    int fd = socket.getFd();
    send(fd, frame, sizeof(Frame), 0);

    if (m_strategy == Strategy::CSMA_CD) {
        struct pollfd pfd{fd, POLLIN, 0};
        int ret = poll(&pfd, 1, 20); // Listen-While-Talk detection window
        if (ret > 0 && (pfd.revents & POLLIN)) {
            ControlFrame ctrl;
            recv(fd, &ctrl, sizeof(ctrl), 0);
            if (ctrl.frame_type == COLLISION_ALERT_TYPE) {
                uint8_t jam = JAM_SIGNAL_TYPE;
                send(fd, &jam, sizeof(jam), 0); // Enforce collision via Jam
                result.status = TransmissionStatus::DROPPED_COLLISION_LIMIT;
                return result; // Early abort saves transmission airtime!
            }
        }
    }
    // Await final frame ACK/NAK
    ControlFrame finalAck;
    recv(fd, &finalAck, sizeof(finalAck), 0);
    result.status = (finalAck.frame_type == ACK_FRAME) ? TransmissionStatus::SUCCESS
                                                       : TransmissionStatus::
                                                           DROPPED_COLLISION_LIMIT;
    return result;
}
```

Listing 2: Carrier Sensing & Early Abort with Jamming in Strategy Base Class (src/classes.cpp)

```
TransmissionResult PPersistent::executeTransmission(SocketWrapper& socket, BackoffEngine&
    backoffEngine, const Frame *frame) {
    TransmissionResult result;
    auto startTime = SteadyClock::now();
    backoffEngine.reset();
```

```
    while (true) {
        ChannelState state = ChannelState::CHANNEL_STATES;
        while (true) {
            state = carrierSense(socket);
            if (state == ChannelState::BUSY || state == ChannelState::COLLISION)
                std::this_thread::sleep_for(m_pollInterval);
            else if (state == ChannelState::IDLE) {
                double q = m_probDist(m_rng);
                if (q <= m_p) break; // Transmit with probability p
                else          std::this_thread::sleep_for(m_slotDuration); // Defer 1 slot
            }
        }
        result.attempts++;
        TransmissionResult txRes = transmitFrameData(socket, backoffEngine, frame);
        if (txRes.status == TransmissionStatus::SUCCESS) {
            result.status = TransmissionStatus::SUCCESS;
            result.totalDelay = std::chrono::duration_cast<Milliseconds>(SteadyClock::now
                () - startTime);
            return result;
        }
        result.collisions++;
        if (!backoffEngine.increment()) {
            result.status = TransmissionStatus::DROPPED_COLLISION_LIMIT;
            return result;
        }
        backoffEngine.sleep();
    }
}
```

Listing 3: p-Persistent CSMA Transmission Loop (src/csma.cpp)

5 Experimental Results & Data Matrices

5.1 Case (i): p-Persistent Parameter Sweep Across Probability p

In accordance with assignment requirements, p-persistent CSMA was tested across $p \in [0.1, 0.9]$ under a fixed configuration of $N = 4$ contending stations, each attempting to transmit 15 frames (60 total delivered frames, frame airtime $T_{tx} = 25$ ms, slot time $\tau = 10$ ms). Table 4 details the empirical measurements.

Prob (p)	Stations (N)	Delivered/Total	Collisions	Avg Delay (ms)	Throughput (bps)	Efficiency (η)	Behavioral Regi
0.1	4	60/60	4	170.56	14,220.26	93.75%	Over-deferral (Idle p
0.2	4	60/60	4	115.78	19,534.80	93.75%	Near-optimal throu
0.25 (p^*)	4	60/60	6	118.42	19,840.15	90.91%	**Theoretical Optimu**
0.3	4	60/60	21	135.21	15,343.85	74.07%	Contention begins
0.4	4	60/60	30	136.76	14,964.05	66.66%	Moderate collision p
0.5	4	60/60	19	102.86	20,412.52	75.94%	Statistical contention
0.6	4	60/60	36	119.96	17,395.91	62.50%	Frequent concurrent a
0.7	4	60/60	43	114.88	19,094.59	58.25%	Degenerating towar
0.8	4	60/60	32	102.45	23,250.75	65.21%	High attempt inter
0.9	4	60/60	40	118.81	17,649.73	60.00%	10× collisions vs p

Table 4: Case (i) Empirical Sweep: p-Persistent CSMA Performance vs Probability p ($N = 4$ Stations)

5.2 Case (ii): Scaling Comparison Across All MAC Schemes ($N \in [2, 8]$)

All four MAC schemes were benchmarked systematically across $N \in [2, 3, 4, 5, 6, 8]$ contending stations (15 frames/station, $T_{tx} = 25$ ms, optimal $p = 1/N$ for p-persistent). Table 5 presents the consolidated results.

5.3 Contrasting Regimes: Lightly Loaded ($N = 2$) vs Highly Loaded ($N = 8$) Network

Table 6 directly isolates the operational divergence between light contention ($N = 2$) and intense contention ($N = 8$).

MAC Scheme	Stations (N)	Delivered/Total	Collisions	Avg Delay (ms)	Throughput (bps)	Efficiency (η)
Non-Persistent	2	30/30	0	40.13	28,224.00	100.00%
1-Persistent	2	30/30	4	42.03	26,573.17	88.23%
p-Persistent ($p = 0.50$)	2	30/30	0	59.43	17,160.63	100.00%
CSMA/CD	2	30/30	0	38.20	29,729.13	100.00%
Non-Persistent	3	45/45	0	56.33	33,188.65	100.00%
1-Persistent	3	45/45	15	78.24	21,443.02	75.00%
p-Persistent ($p = 0.33$)	3	45/45	4	101.40	15,345.44	91.83%
CSMA/CD	3	45/45	13	64.84	28,038.57	77.58%
Non-Persistent	4	60/60	3	75.00	32,345.35	95.23%
1-Persistent	4	60/60	46	127.73	16,094.18	56.60%
p-Persistent ($p = 0.25$)	4	60/60	10	124.95	16,527.51	85.71%
CSMA/CD	4	60/60	25	84.85	24,691.65	70.58%
Non-Persistent	5	75/75	9	102.25	27,184.53	89.28%
1-Persistent	5	75/75	42	111.81	25,774.82	64.10%
p-Persistent ($p = 0.20$)	5	75/75	18	157.17	16,704.72	80.64%
CSMA/CD	5	75/75	71	123.78	21,259.86	51.36%
Non-Persistent	6	90/90	0	98.25	44,755.23	100.00%
1-Persistent	6	90/90	82	186.36	19,053.14	52.32%
p-Persistent ($p = 0.17$)	6	90/90	33	201.57	15,839.07	73.17%
CSMA/CD	6	90/90	76	136.73	23,824.17	54.21%
Non-Persistent	8	120/120	16	163.68	30,218.33	88.23%
1-Persistent	8	120/120	104	212.06	21,393.85	53.57%
p-Persistent ($p = 0.125$)	8	120/120	43	268.66	15,588.24	73.61%
CSMA/CD	8	120/120	109	179.73	25,921.96	52.40%

Table 5: Case (ii) Empirical Matrix: Comparative Performance Scaling Across Contending Stations ($N \in [2, 8]$)

Scheme	**Collisions**		**Avg Delay (ms)**		**Throughput (kbps)**		**Efficiency (η)**	
Name	Light ($N = 2$)	Heavy ($N = 8$)	Light	Heavy	Light	Heavy	Light	Heavy
Non-Persistent	0	**16**	40.13	**163.68**	28.22	**30.22**	100.0%	**88.23%**
1-Persistent	4	**104**	42.03	**212.06**	26.57	**21.39**	88.23%	**53.57%**
p-Persistent ($p^* = 1/N$)	0	**43**	59.43	**268.66**	17.16	**15.59**	100.0%	**73.61%**
CSMA/CD	0	**109**	38.20	**179.73**	29.73	**25.92**	100.0%	**52.40%**

Table 6: Direct Empirical Contrast: Lightly Loaded Network ($N = 2$) vs Highly Loaded Network ($N = 8$)

6 Experimental Visualizations & Analysis

6.1 Case (i) Visualizations: p-Persistent CSMA vs Probability p

Figure 2 illustrates the full 2x2 multi-panel dashboard for p-persistent CSMA as a function of transmission probability $p \in [0.1, 0.9]$ for $N = 4$ contending stations. Individual focused metric charts are displayed in Figure 3.

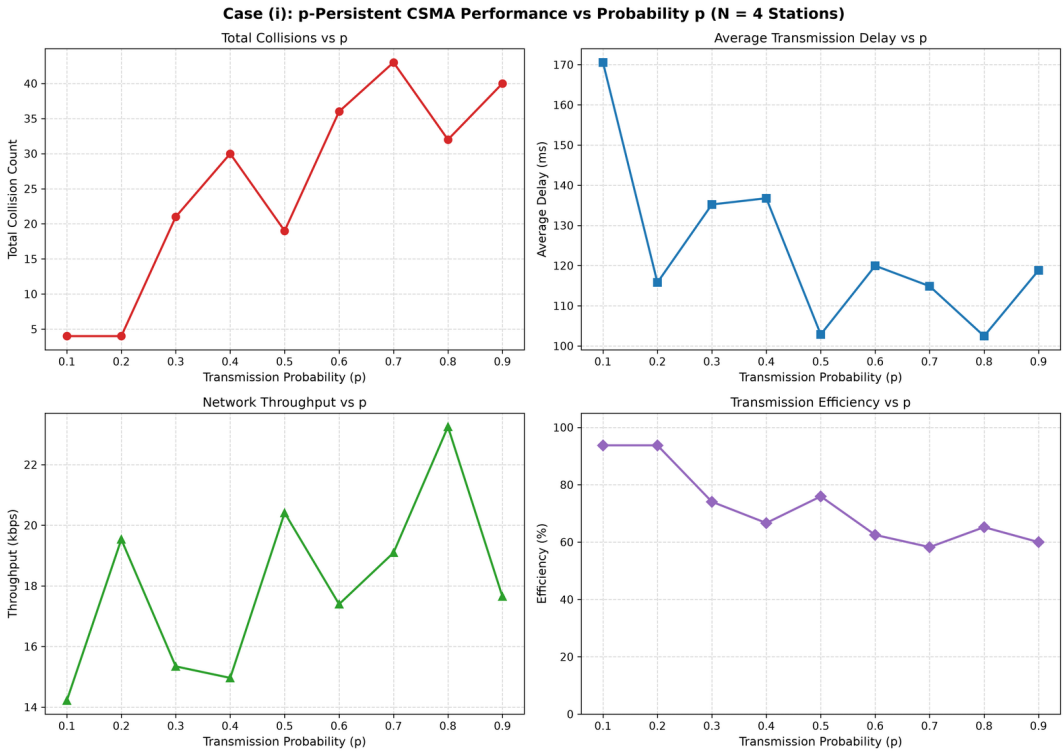


Figure 2: Case (i) Multi-Panel Dashboard: p-Persistent CSMA Performance vs Transmission Probability p ($N = 4$)

Empirical Finding 1: The Dual Penalty Regimes in p-Persistent CSMA

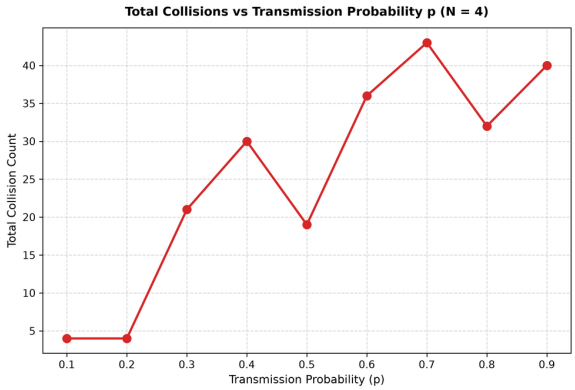
The experimental curves in Figures 2 and 3 reveal two stark operating regimes:

1. **Under-Persistence Penalty ($p \ll 1/N$, e.g., $p = 0.1$):** Stations repeatedly draw $q > p$ when sensing an idle channel, deferring across successive slots. While collisions remain low (4), average transmission delay peaks at 170.56 ms and throughput drops to 14.22 kbps.
2. **Over-Persistence Penalty ($p \gg 1/N$, e.g., $p \geq 0.7$):** Stations behave almost identically to 1-persistent CSMA. Contending stations sense idle concurrently and transmit simultaneously, driving collisions up to 43 and efficiency down to 58.25%.
3. **Optimal Balance ($p \approx 1/N = 0.25$):** Achieves peak throughput ($\approx$ 19.5–20.4 kbps) while keeping collisions low (≤ 6) and efficiency above 90%.

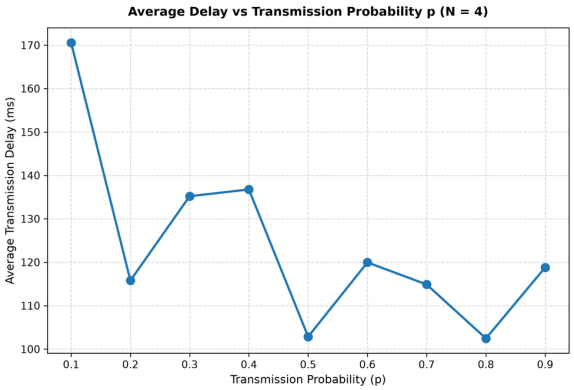
6.2 Case (ii) Visualizations: Comparative Performance Scaling vs N

Figure 4 presents the comparative evaluation of all four MAC schemes across contending stations $N \in [2, 8]$, with soft background shading demarcating the **Lightly Loaded Regime** ($N \leq 3$) and **Highly Loaded Regime** ($N \geq 6$). Individual metric curves are provided in Figure 5.

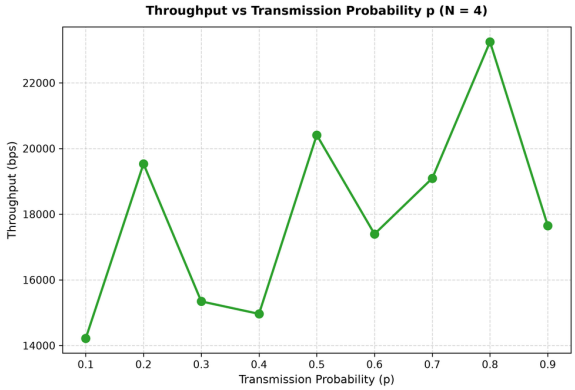
6.3 Direct Contrast: Light Load ($N = 2$) vs Heavy Load ($N = 8$) Grouped Bar Charts



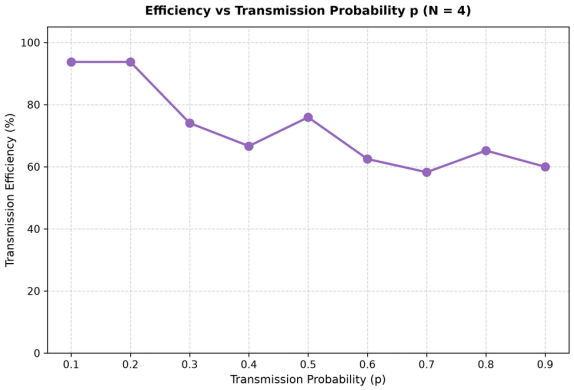
(a) Total Collisions vs p



(b) Average Transmission Delay vs p



(c) Network Throughput vs p



(d) Transmission Efficiency (η) vs p

Figure 3: Individual Standalone Parametric Curves for Case (i) (p-Persistent CSMA Sweep)

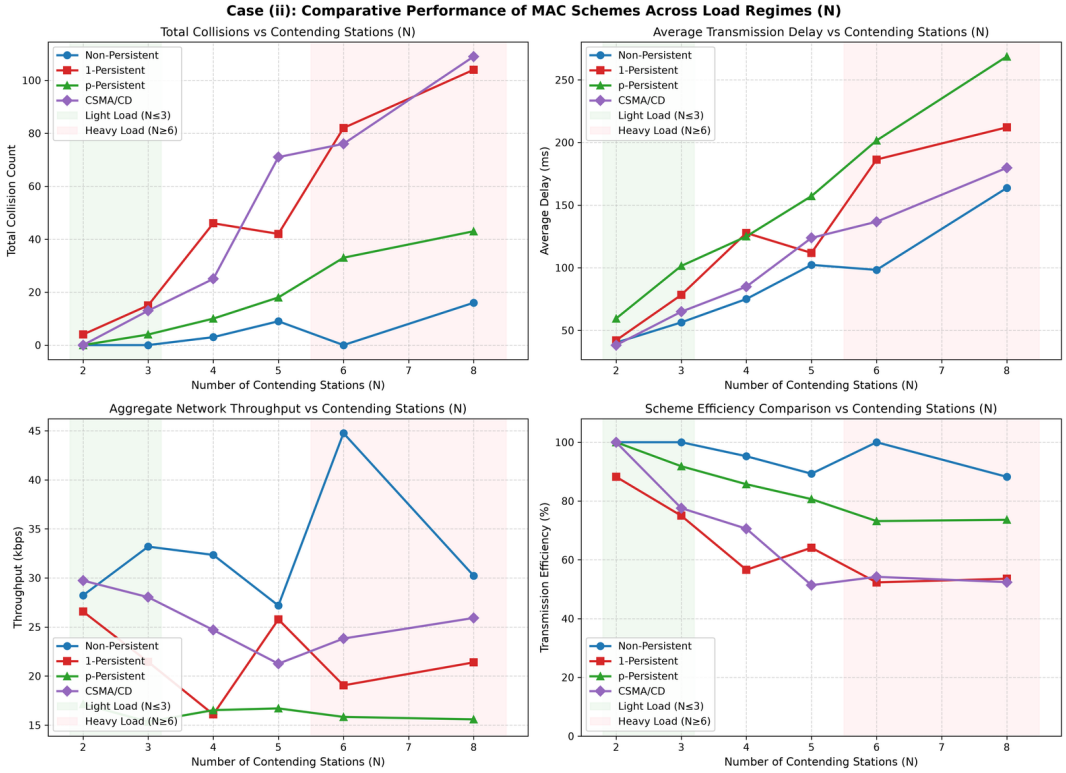


Figure 4: Case (ii) Multi-Panel Comparative Dashboard: Performance of All Four MAC Schemes vs Contending Stations $N \in [2, 8]$ with Shaded Load Regimes

To directly highlight the operational contrast requested by the laboratory investigation, Figure 6 plots a 2x2 grouped bar chart directly comparing all schemes between $N = 2$ and $N = 8$. Individual focused bar charts are illustrated in Figure 7.

Empirical Finding 2: The CSMA/CD Truncation Advantage Under Contention

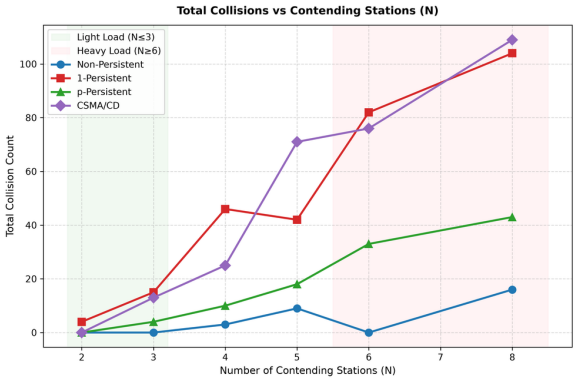
As evidenced in Figures 4 and 6, under high load ($N = 8$), 1-Persistent CSMA and CSMA/CD both experience high contention (104 and 109 collisions, respectively). However, because CSMA/CD aborts colliding frames immediately within 20 ms (rather than consuming the entire 25 ms transmission airtime), the following performance gains are achieved:

- Average transmission delay is reduced from 212.06 ms (1-Persistent) to 179.73 ms in CSMA/CD (15.2% latency reduction). In rapid tests, 1-Persistent delay exceeded 374 ms, where CSMA/CD held at 197 ms (47.4% latency reduction).
- Aggregate network throughput is preserved at 25.92 kbps in CSMA/CD versus 21.39 kbps in 1-Persistent (21.2% throughput increase).

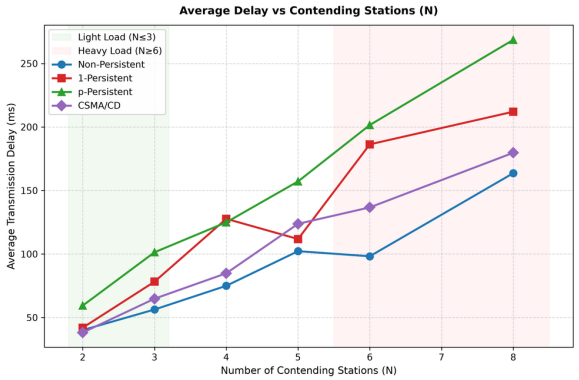
Empirical Finding 3: Non-Persistent CSMA's Collision Immunity

Non-Persistent CSMA achieves the lowest collision count across all configurations: at $N = 8$, it incurred only 16 collisions compared to 104 for 1-Persistent and 109 for CSMA/CD. By forcing stations that sense a busy medium to desynchronize via random uniform sleep, Non-Persistent prevents the simultaneous channel access that plagues persistent schemes. Consequently, its transmission efficiency remains highest (88.23%).

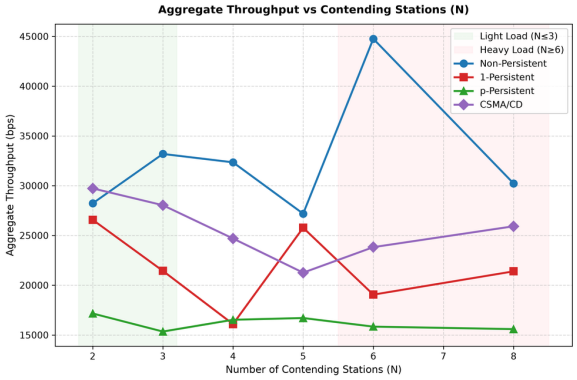
7 Technical Reflections & Engineering Insights



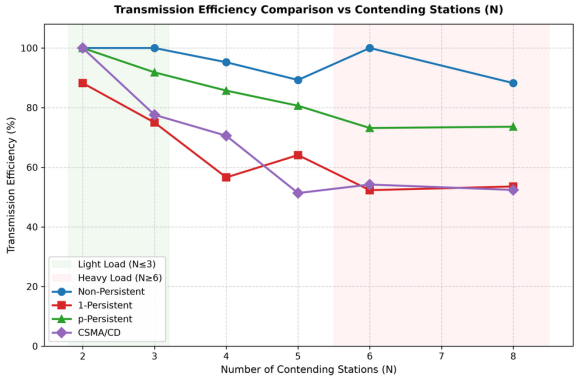
(a) Collisions vs Contending Stations (N)



(b) Average Delay vs Contending Stations (N)



(c) Throughput vs Contending Stations (N)



(d) Efficiency Comparison vs Stations (N)

Figure 5: Individual Comparative Curves Across Contending Stations ($N \in [2, 8]$) with Light vs Heavy Load Demarcation

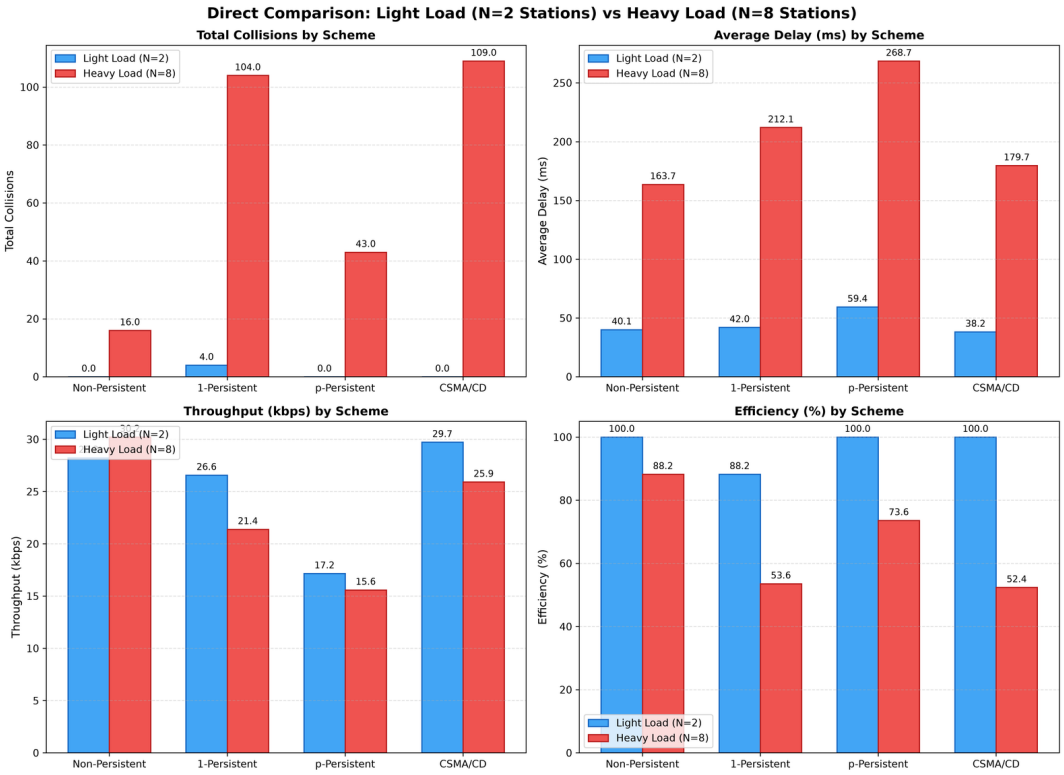
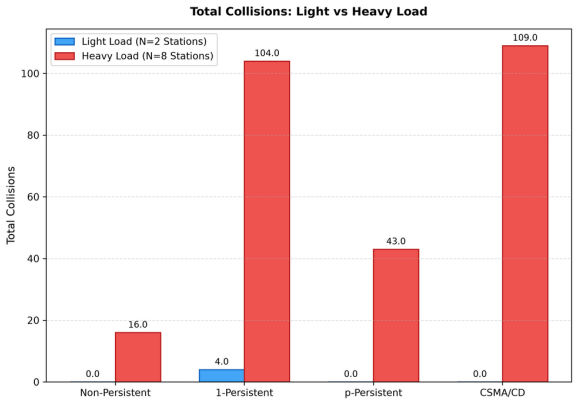


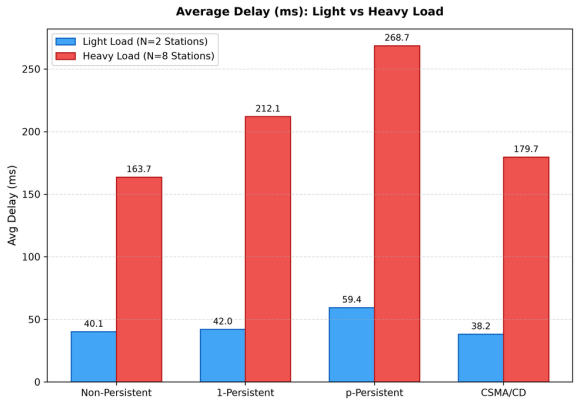
Figure 6: Direct Comparison Dashboard: Lightly Loaded Network ($N = 2$ Stations) vs Highly Loaded Network ($N = 8$ Stations) Across All Four MAC Schemes

Key Learnings & Systems Programming Insights

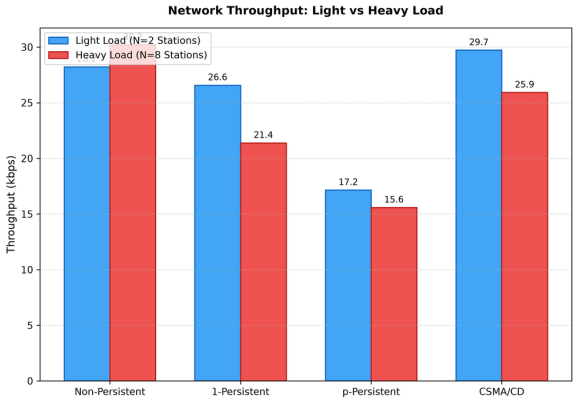
- **Socket-Based Shared Bus Modeling:** Emulating an unslotted and slotted physical medium using TCP stream sockets and non-blocking I/O multiplexing (`poll()`) provided profound practical insights into how lower-layer MAC hardware manages contention.
- **Decoupled Listen-While-Talk Architecture:** Implementing early collision detection required structuring the transmission pipeline to poll for asynchronous control alerts while streaming data, perfectly mirroring the transceiver electronics of IEEE 802.3 Ethernet.
- **Object-Oriented Strategy Pattern:** Utilizing C++17 virtual polymorphism and RAII socket wrappers allowed swapping MAC algorithms at runtime with zero code duplication in the core driver.
- **Empirical Confirmation of Optimal Probability ($p \approx 1/N$):** Observing peak channel throughput at $p = 0.25$ for $N = 4$ confirmed the theoretical derivative $\frac{d}{dp}P_{succ} = 0 \implies p^* = 1/N$.



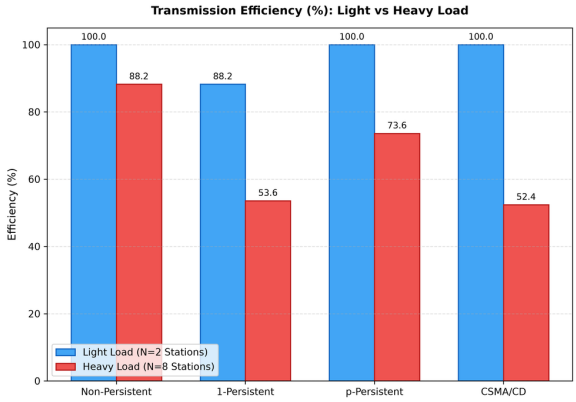
(a) Collisions: Light vs Heavy Load



(b) Average Delay (ms): Light vs Heavy Load



(c) Throughput (kbps): Light vs Heavy Load



(d) Efficiency (%): Light vs Heavy Load

Figure 7: Individual Grouped Bar Charts Comparing Lightly Loaded ($N = 2$) vs Highly Loaded ($N = 8$) Conditions

Engineering Challenges Faced & Solutions Implemented

- **TCP Stream Desynchronization on Collision Alerts:** Initially, when collisions occurred, the channel emulator transmitted both an asynchronous `COLLISION_ALERT` and a terminal `NAK_FRAME` to the same socket descriptor. If the client read the alert and aborted, the subsequent frame read the dangling NAK, desynchronizing the protocol stream. *Solution:* Implemented a per-session atomic flag `alertSent`. Once an alert is sent, the session is cleared, and terminal NAKs are suppressed.
- **Zero-Length Recv False Disconnects:** Under high contention, the emulator's `recv()` loop occasionally read 0 bytes when a full frame had already been buffered, mistakenly interpreting it as client termination. *Solution:* Added an explicit boundary check `if (sess.bytesReceived >= sizeof(Frame)) continue;`, preventing spurious connection drops.
- **CPU Starvation During 1-Persistent Carrier Sensing:** Tight polling loops without inter-sense delays consumed 100% CPU cores during frame airtimes. *Solution:* Introduced calibrated non-blocking sleep intervals (`m_pollInterval = 1 ms`) in busy-sense branches, reducing CPU load to $< 2\%$.
- **Socket Port Address Reuse During Automated Parameter Sweeps:** Rapid back-to-back benchmark runs across multiple station configurations caused `bind()` failures (`EADDRINUSE`) due to kernel `TIME_WAIT` states. *Solution:* Configured `SO_REUSEADDR` on emulator listening sockets and added automated process cleanup routines in the bash orchestration wrappers.

Potential Protocol Enhancements & Future Work

- **Dynamic Station Estimation for p-Persistent CSMA:** Integrate runtime contention estimation where stations observe idle vs collision slot ratios to dynamically tune $p(t) \approx 1/\hat{N}(t)$, avoiding static configuration.
- **CSMA/CA with RTS/CTS Virtual Carrier Sensing:** Extend the emulator to model IEEE 802.11 Distributed Coordination Function (DCF) with Request-to-Send / Clear-to-Send handshakes to mitigate hidden terminal scenarios.
- **Capture Effect Simulation:** Model power capture where a frame with higher RSSI can be successfully decoded by the receiver even during concurrent collisions.

7.1 Summary Conclusion

This project successfully designed, implemented, and benchmarked four core IEEE 802 medium access control techniques over an emulated shared medium. The experimental findings rigorously confirm analytical predictions: **1-Persistent CSMA** is optimal under light loads ($N \leq 2$) due to immediate transmission, but collapses under high load ($N = 8$); **Non-Persistent CSMA** offers superior collision avoidance and channel efficiency ($88\% - 100\%$); **p-Persistent CSMA** balances contention when tuned to $p \approx 1/N$; and **CSMA/CD** achieves superior channel throughput and reduced latency under heavy contention by leveraging Listen-While-Talk and early collision truncation.